

BILKENT UNIVERSITY
Department of Electrical and Electronics Engineering
EEE102 Introduction to Digital Circuit Design
MidTerm Exam I
14-03-2005
Duration 110 minutes

Surname: _____

Name: _____

ID-Number: _____

Signature: _____

There are 7 questions of different weights. Solve all. Do not detach pages.

Q1	
Q2	
Q3	
Q4	
Q5	
Q6	
Q7	
TOTAL	

1. (16 points)
- a) Show that $A \cdot B = A \cdot C$ does not imply that $B = C$.
 - b) Show that NAND and NOR operators are not associative.
 - c) Given that $X \cdot Y' + X' \cdot Y = Z$, show that $X \cdot Z' + X' \cdot Z = Y$.
 - d) Convert the expression $\{[(A+B+A' \cdot C') \cdot C+D]' + A \cdot B'\}$ into sum of minterms.

2. (12 points) Find all minimal sum expressions and all minimal product expressions for

$$F = \sum_{A,B,C,D} (0, 2, 3, 4, 8, 10, 11, 12, 14, 15).$$

(Draw Karnaugh Maps, indicate prime implicants and distinguished 1s and 0s).

3. (16 points) Find all minimal sum expressions and all minimal product expressions for

$$F = \sum_{A,B,C,D} (0, 2, 3, 4, 12, 13) + d(5, 7, 10, 11, 14).$$

Determine which of these minimal expressions are equivalent. Explain why.
(Draw Karnaugh Maps, indicate prime implicants and distinguished 1s and 0s).

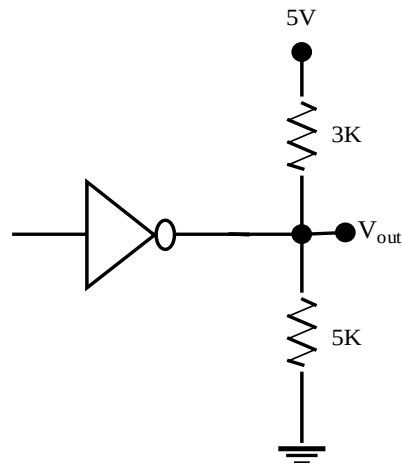
4. (12 points) Find all minimal sum expressions for

$$F = \prod_{A,B,C,D} (1, 3, 6, 7, 9, 11, 12, 13, 14, 15).$$

Also find all minimal product expressions for F by using the methodology of finding minimal sums.

(Draw Karnaugh Maps, indicate prime implicants and distinguished 1s and 0s).

5. (16 points) A CMOS inverter working from 5V supply has a resistive load as shown below.



Suppose this CMOS inverter has the following specifications:

$$V_{IHmin} = 3.5 \text{ V}, V_{ILmax} = 1.5 \text{ V}, I_{ILmax} = -10 \mu\text{A}, I_{IHmax} = 15 \mu\text{A}.$$

The ON and OFF resistances of the NMOS and PMOS transistors are $1\text{K}\Omega$ and $1000\text{K}\Omega$ respectively.

- Find V_{out} for when it is HIGH and also for when it is LOW.
- How many additional CMOS inverters of the same type can be connected to V_{out} ?

6. (12 points) Draw the internal circuit of a CMOS circuit (using NMOS and PMOS transistors) which has the logic function $F = A' \cdot B' + C' \cdot D'$.

7. (16 points) The rectangular block in the below drawing represents a logic circuit (logic block). Design this logic circuit so that if EN is 0 then F is in Hi-Z state and if EN is 1 then $F = A'$. In other words, you are to design the inside of the logic block so that the whole circuit becomes a three-state inverter. Make your design using gates (not transistors). Draw the whole circuit.

